# Week 12

## Reproducibility & Data Management

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# Check-ins

- Email me your current advisor situationship
  - Even if you have an advisor
  - We have 3-4 weeks left
- Spreadsheet here

# Data Documentation

- It is important to have documentation of your project, especially when collaborating with others
- One of my favorite resources
  - Data Management in Large-Scale Education Research
    - Level of aggregation and cleaning before sharing
- If using advisor's data, this becomes even more important since there could be legal issues if mistakes are made, like data leakage

# Documentation

- Variables
  - Make sure data types are stated
  - Variable types
  - Operationalization of variables
  - Units of measurement
  - Missing data
    - Missing data rules
    - How data will be imputed

# Documentation

- Descriptions of workflows
- Access permissions fr specific variables
  - Unit of analysis level
- Staying within the scope of regulating organizations (HIPAA)

# Codebook

- I like to have a specific file that has more information about each of my variables
- Codebooks are especially important for longitudinal data/cohort design tables
  - Makes data manipulation consistent across different data collection points
  - Also makes it transparent when mistakes are made
    - Coding of variables inconsistently happens
- Putting it in a Jupyter notebook/Quarto document allows for more organization

# Codebook Example

- We'll focus on the CESD, a common measure for depressive symptoms
  - You can follow along here
- What to include
  - Name of variable(s)
    - variable name in dataset
  - Data type
  - Variable type
    - Recommendations for transformations
    - Scoring to make more sense –> new variable name(s)
  - Scale/Range of variable
  - Some people include descriptives (M, SD, range, etc.)
  - Skip logic in questions
  - Logical ways to get composite scores
    - Positive affect & Depressive symptoms
  - Rules for missing data
  - Amount of missing data
  - References (bibtex can be used in documents)
    - Articles of interest

# Codebook

- Include relevant information
  - Visualize missing data
  - Visualize distribution(s)
  - Render table(s) for potential problematic values
  - Define cutoffs from previous literature
  - Recoding based on

# Reproducibility & Open Source

- Having your code in a public space shows reproducibility
- Public repos showcase
  - Your code changes through commit history
  - Decision making
  - If you try to hide something
    - Setting a seed/state for analyses and then changing it
- Sharing everything about your project
  - May not be feasible for some of your projects
- Opens the door for collaborations with outside researchers

# Open Science

- Making research equitable by making everything in the research process more accessible
- Openly sharing
  - Software and code
  - Data*
  - Published research articles
  - APIs
- Makes it more reproducible too
  - When others have your code and data, they can double check your work to make sure the findings are accurate
- **Data may not always be completely raw**
  - Still need to protect identities for participants/subjects

# Open Science

- Making science open is expensive
  - To make a peer-reviewed article open access, it often costs thousands of dollars
  - Publishers guilt trip authors into wanting open access
  - Often negotiations between universities and publishers on price
- Publishers want $, authors want to expand on literature

# Git & GitHub

- Raise your hand if you are familiar with Git and GitHub/GitLab
- GitHub Desktop
- Github Education
- Important considerations
  - GitHub information on choosing a license
    - Choose the appropriate license
  - Make sure your `.gitignore` file is updated and correctly ignoring any sensitive data/information

# Collaboration

- Having documentation standards
  - This includes writing out interpretable comments in scripts
  - Documents (Readme files) with more information about each directory in your project
  - How to make pull requests
    - Reviewing pull requests
  - Incorporating tests to make sure code works appropriately and accurately

# Project/Package Management

- Decide what you want to use
  - virtual environment
- uv
  - Sync with collaborators
- Other options

# Docker

- Getting started with Docker
- Docker containers for Python
- Docker sandbox
- Docker allows you to containerize your project that will run on any machine
- If interested in putting your Docker image on the Docker hub, you can save your project as a docker image then publish it
  - This will stop or at least lessen issues with dependencies in your project

# HPC & Savio

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- Apptainer for a secure container for HPC
- Python in Savio

# Examples of GitHub Repos

- JP's GitHub Repo
- Previous year's data science honors thesis
- Another data science honors thesis
- Population Inference Using Savio

# Presentations

- Everyone gets ~8 minutes
- Feedback will be sent over bcourses
  - Everyone is expected to respond to their comments
- You'll need to cover:
  - Who your advisor is
  - Why something is a problem
  - Small amount of literature
  - RQs/Hypotheses
  - Methods
    - What data you'll use
    - How your data is/was collected
    - Analytic plan
    - How you'll analyze data (locally, find resources, etc.)